

Buoyancy fluxes of sea-ice leads in the Arctic

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Key Points:

- Buoyancy fluxes in Linear Kinematic Features vary seasonally
- Buoyancy fluxes in Linear Kinematic Features can be 20-30% larger than in the ice-pack.
- The presence of Linear Kinematic Features has the potential transform water masses in the ocean surface.

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Abstract

The Arctic sea-ice, in particular the ice-pack, acts as an insulator between the atmosphere and the ocean below. Linear kinematic features (LKF), or leads, common in Arctic sea ice serves as openings that allow exchanges between ocean and atmosphere, for example: heat, salt, moisture, and gases. Therefore, leads can influence heat exchanges, ice melt or growth, ocean dynamics, and the atmospheric boundary layer. The scale of leads varies from a few meters to kilometers in width, and from few kilometers up to hundreds of kilometers in length. Their formation is driven by the winds and ocean currents. Existing literature has suggested that leads play a significant role in the heat budget of the surface of the ocean and likely impact water mass transformation within the Arctic. However, due to the lack of observations and high resolution models resolving LKFs, an estimate of the magnitude of the impact in the ocean has been elusive. Here, we explore the magnitude and impact in the ocean surface caused by the LKFs using SEDNA, a high resolution pan-Arctic hindcast ($1/60^\circ$; NEMO + SI3 configuration). We show that despite the LKF only account for a small area of the sea-ice cover (5%), they contribute up to 20-30% more buoyancy than the ice-pack at the surface. Furthermore, of the total surface water mass transformation in the Arctic ice cover, LKFs explain approximately 10% of it. LKFs make a significant contribution despite their small area coverage; however, the presented estimate shows a large local impact but to a lesser degree in the Arctic Ocean than previously hypothesized.

1 Introduction

The Arctic sea-ice plays a crucial role in regulating Earth's climate by serving as a natural insulator between the atmosphere and the ocean below (Wettlaufer et al., 1997; Untersteiner, 1961). This insulator property acts as a thermal barrier, limiting heat loss from the ocean to the atmosphere and minimizing evaporation, thereby it maintains the stability of the polar environment (Ruffieux et al., 1995). An ubiquitous feature of the Arctic sea ice is the formation and persistence of linear kinematic features (LKF), commonly known as leads. These features are elongated openings of the sea ice cover that expose the ocean surface to the atmosphere (Lüpkes et al., 2008).

Leads vary in scale from a few meters to kilometers in width, and span distances of a few kilometers up to hundreds of kilometers (Tschudi et al., 1998). Their formation mechanisms are diverse, ranging from mechanical to thermodynamic processes (Ram-

pal et al., 2016; Hutchings et al., 2005). Mechanically, sea ice deformation and fracturing are caused by winds and ocean currents (Richter-Menge et al., 2002). Winds are thought to be the main forcing of the sea ice deformation, where persistent winds stress result in sea ice fracturing, and eventually leads (Linow & Dierking, 2017; Hutter et al., 2018). Although ocean currents have a smaller contribution to the sea ice deformation compared to the wind forcing, ocean drag at the bottom of the ice acts to balance the wind-induced sea ice velocities (Weiss & Marsan, 2004). Additionally, observations show that ocean currents may play a bigger role in dictating the fracturing patterns of sea ice in regions with strong currents and bath features (Willmes et al., 2023). Alongside these mechanical forces, thermodynamic processes such as a rapid variation of the air or ocean temperature can result in the fracturing of sea ice (Lewis, 1998). Once a lead is formed, warm air temperatures can instigate localized melting maintaining the lead open. Conversely, during colder periods, rapid freezing facilitate the close a lead. The intricate interplay between these thermodynamic processes and dynamic forces dictates the evolution and persistence of leads.

Leads are crucial since they act as gateways between the atmosphere and the ocean, exchanging heat, moisture, and gases (Michaelis & Lüpkes, 2022; Andreas et al., 2002). Depending on the season and the atmospheric forcing, this direct contact between the ocean water and atmosphere can result in locally intensified heat fluxes at the ocean-atmosphere interphase (Marcq & Weiss, 2012; Maykut, 1986), localized ice melt or growth, and modify the ocean dynamics (Bourgault et al., 2020). For example, in winter heat exchange from the ocean to the cold atmosphere, in addition to ice freezing within the leads, result in local convective processes in both the ocean and atmosphere (Lüpkes et al., 2008; McPhee et al., 2005). A change in the surface temperature and salinity due to the heat and salt flux at the ocean-ice -atmosphere interfaces transform locally the surface water masses (Walsh, 1982). Leads have a undeniable local impact in the ocean and it has been hypothesized that leads may play a significant role in the Arctic by modulating the stratification, altering ocean circulation and transformation of water masses (Reiser et al., 2020; Smith et al., 1990). However, the magnitude and influence of the LKFs compared to the surrounding ice pack in the Arctic basin is yet to be quantified.

To investigate and quantify the impact of the LKFs in the Arctic ocean, we use a really high resolution pan-Arctic hindcast; SEDNA (Talandier & Lique, 2023). Then, we compute a mask of the linear kinematic features during the 2014 year, as well as the

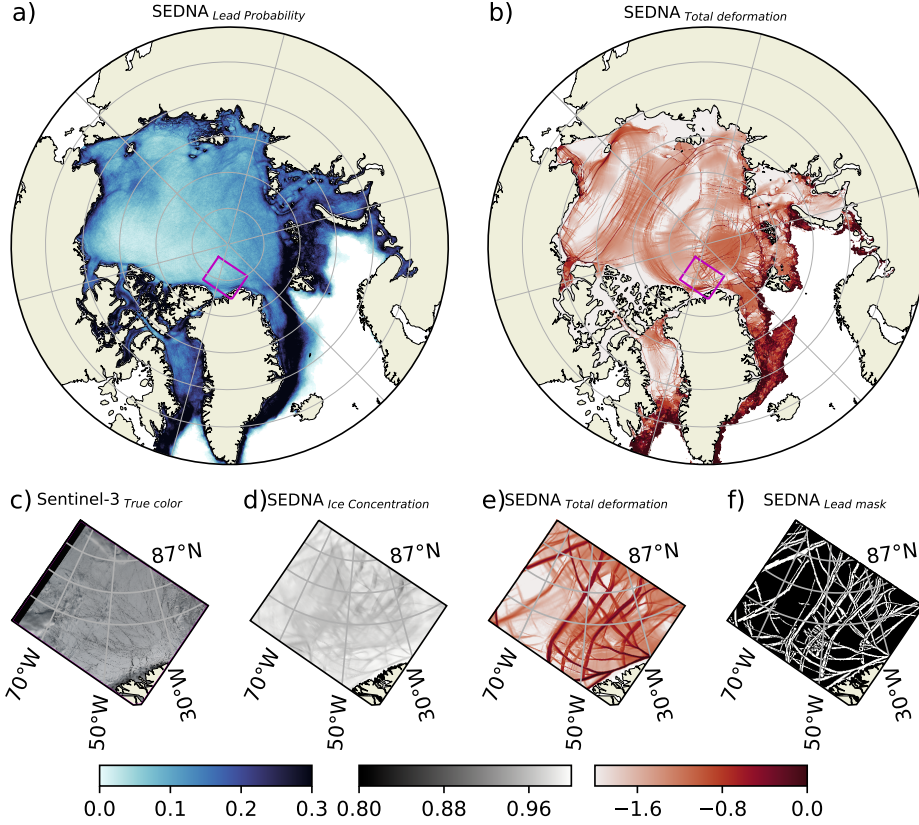


Figure 1. Linear kinematic features from SEDNA. a) Linear kinematic mean frequency during winter between 2011 and 2015. b) Snapshot of the total deformation for the XX of XX. c) True color image from Sentinel 3 on the 12th of April 2022 for the magenta area in panel a. d) Mean ice concentration from SEDNA (800m resolution) on the 21st of April 2014. e) Total deformation used to identifies the linear kinematic features. f) Identified linear kinematic features on the 21st of April 2014.

buoyancy fluxes and water mass transformation for the identified LKFs and compare them to the ice pack (See Methods for details of the calculations and numerical model).

Observation vs Models

Climate models and high resolution sea ice models are unable to reproduce leads due to their small footprint in the Arctic (Wang et al., 2016). SEDNA is capable to overcome this limitation, due to its spatial resolution of $\sim 800m$ on average in the Arctic. The climatology of the winter lead probability is shown in Figure 1a. There are several distinct spatial patterns that match those from observations (Willmes et al., 2023). For

example, we observe increased lead frequency in the shelves, along bathymetric features and in regions with large kinetic energy. Several hotspots can be identified along the sea shelves and in the Chukchi Sea, highlighting the model's ability to reproduce the LKF dynamics.

SEDNA is capable of reproducing the overall pattern of the observed winter lead distribution in the Arctic shelves, the Greenland Sea, Barents Sea, Kara Sea and Laptev Sea, however, observations show a prominent signature in the Beaufort Sea that is not captured in our simulation. A snapshot of the total deformation is shown in Figure 1b, where the local maxima in the deformation correspond to the linear kinematic features identified by the algorithm described in the Methods section. A closer examination, focusing north of Greenland in panels c, d, e, and f of Figure 1 reveals similarities between the spatial pattern of typically observed leads on the 12th of April 2022 from a true color satellite image (Fig. 1c) and the ice concentration of SEDNA for the 21st of April 2014. Note that the sea ice concentration in the identified leads shown in Fig.1e-f remains higher than 80% concentration, this is a consequence of the visco-plastic nature of the sea ice model and the daily averaged output. Furthermore, in our simulation, but in fact in all numerical models using a visco-plastic rheology it has been observed that they do not reproduce the typical fracturing angles as in observations (). Despite this, the important similarities between the observations and model lead probability gives us confidence to estimate the impact of leads in the ocean.

In situ observations have shown that LKF in winter can have heat fluxes up to two orders of magnitude larger than the surrounding ice covered ocean. Presumably impacting the buoyancy flux and the circulation of the Arctic Ocean. Figure 2a and e show snapshots of the buoyancy flux, where we can observe the typical patterns of the Arctic. In winter (Fig. 2a) we observe an overall negative buoyancy flux consequence of the freezing of sea ice and thus brine rejection, meanwhile, in summer an overall positive buoyancy flux occurs due to the melting and freshening of the Arctic (Fig. 2b). Overall, in these two maps we observe that inside the ice covered region (magenta contour), the fluxes are spatially homogeneous with exception of the MIZ and the LKFs. For the latter features, we observe that linear kinematic features have larger buoyancy fluxes than their surroundings.

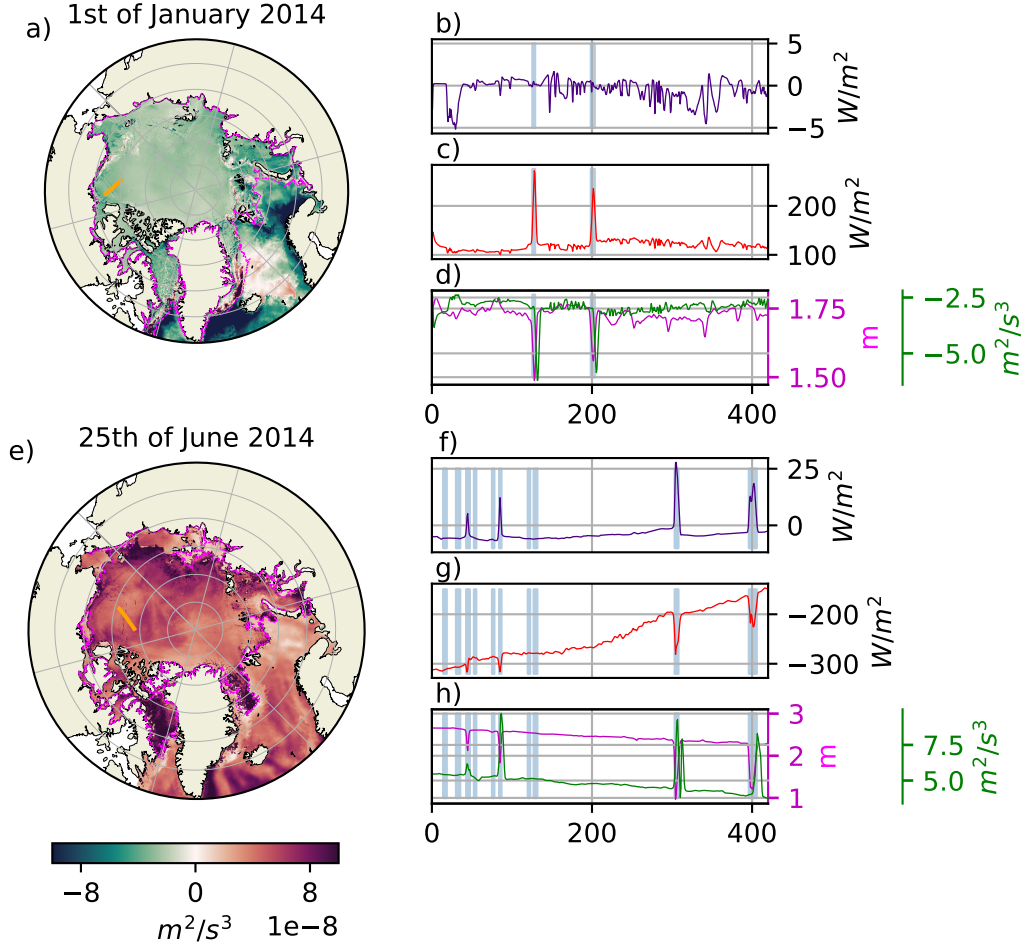


Figure 2. Maps of buoyancy flux and transects of downward heat flux, freshwater heat-equivalent flux, ice thickness, and mixed layer depth for the 1st of January and 25th of June 2014. a) Buoyancy flux in Winter the 1st of January 2014. Transect on the 1st of January 2014 of b) downward heat flux, c) freshwater heat-equivalent flux, d) ice thickness and mixed layer depth. e) Buoyancy flux in Summer the 30th of June 2014. Transect on the 25th of June 2014 of b) downward heat flux, c) freshwater heat-equivalent flux, d) ice thickness (magenta) and buoyancy flux (green). Note that the transect location and orientation changes between the winter and summer dates. Magenta contour in panels a and e correspond to the 0% concentration contour of sea ice. Vertical blue bars shows the mask of LKFs in the transect.

In figure 2, we plot two transects on the 1st of January 2014 and 30th of June 2014 showing the heat flux (panels b, f), freshwater heat-equivalent flux (panels c and g), and ice thickness and buoyancy flux (panels d and h). Examining the 1st of January 2014, we observe a small heat flux where the LKFs (Fig.2 b); blue shaded areas) do not impact the heat flux at the ocean-atmosphere interface. In contrast, the salt flux (Fig.2c) show two peaks with twice the magnitude of the surrounding ice cover collocated with the identified LKFs. A positive freshwater flux corresponds to a salinification of the ocean surface due to sea ice growth and a negative buoyancy flux that increases the density of the water at the surface (Fig.2d). As expected, there is an decrease in ice thickness where the LKFs are identified. On the 25th of June (Fig.2f), we observe an increase in the heat fluxes within some of the identified LKFs to up to $\approx 25W/m^2$. Meanwhile, the freshwater flux show a decrease within the LKFs suggesting a freshening of the surface due to sea ice melt that is captured by the positive buoyancy flux decreasing the density at the surface (Fig.2g and h). Analogously to the winter snapshot, we note a decrease in the ice thickness of more than 1m within the LKFs. It is important to note that some peaks in the buoyancy flux align with the edge of the detected leads likely due to the enhanced refreezing or melting that occur at the edge of the LKFs. Furthermore, not all the identified LKFs show important ocean-atmosphere fluxes, for these features we explored snapshots of the simulation a few days later and noted that the LKFs are starting to open and thus the ocean and ice still behave as within the ice pack (**Check this idea?**).

As in observations, we detect heat fluxes larger in the LKFs than in the surrounding ice pack in both summer and winter, however, the freshwater heat-equivalent flux due to melting and freezing is significantly larger than the heat fluxes associated to the LKFs. By identifying the a daily mask of the LKFs, we explore in the next section the evolution of the buoyancy fluxes over the full 2014 year.

Impact of LKF in the ocean buoyancy

We now consider the spatially averaged impact of LKFs in the ocean buoyancy. We start by look at the percentage area covered by 3 distinct masks of the sea ice covered Arctic Ocean; i) the daily identified LKF, ii) the marginal ice zone (15-80% sea ice concentration), and iii) the ice pack (80-100% sea ice concentration). The area covered by the ice pack shows a seasonal cycle, during summer the percentage of ice cover area drops

from $\approx 90\%$ to 50% . In contrast, the marginal ice zone during summer increases from 10% to $\approx 50\%$. Meanwhile, the percentage of area covered by LKFs never exceeds more than 5% cover of the sea ice cover Arctic and it show a seasonal cycle with a maxima in the winter season due to the stronger storms capable to break the ice pack in flows

Check this!

To demonstrate the role of leads in buoyancy flux, we multiply the buoyancy flux and area weighted them by the respective masks of the sea ice cover; the MIZ, the ice pack, and the LKF. The buoyancy flux show an important seasonal cycle, where positive values correspond to melting season of sea ice, thus forming lighter waters in the ocean surface, and negative values correspond to the freezing season of sea ice, thus forming denser waters. The buoyancy flux in the MIZ forms generally lighter waters (Fig.3b), this suggests that in the MIZ there is a preferential melting of sea ice melt, where both the haline and thermal components of the buoyancy flux act in the lightening of the ocean surface (Fig.3c and d). Over the seasonal cycle, the pack ice changes sign from winter to summer (Fig.3b), this sign change corresponds to a shift from ice formation in winter (formation of denser waters) to ice melting in summer (formation of lighter waters), where the primary contribution to the buoyancy flux arises from the haline buoyancy component. Analogous to the ice pack, the buoyancy in the LKF have a seasonal cycle and it shows a similar temporal evolution as the ice pack. Although LKFs cover less than 5% of the total Arctic ice cover area, over certain periods of time the LKF buoyancy flux can be 10% - 20% larger than that of the ice pack.

To further quantify the role of the ice pack and the LKF on the on the overturning circulation of the Arctic, we use the water mass transformation framework (Walsh, 1982). This approach gives us information on the formation of lighter or denser water due to changes in the surface fluxes. In agreement with previous studies, the surface freshwater flux dominates the water mass transformation of the Arctic. The water mass transformation of the ice pack is characterized by a broad region of strong positive transformation in winter (losing buoyancy) in the density range of $25 < \gamma_n < 27.7$, reaching a maximum of -8 Sv, and a region of negative transformation in summer (gaining buoyancy).

Figure 4

- TODO: Add units below colorbar.

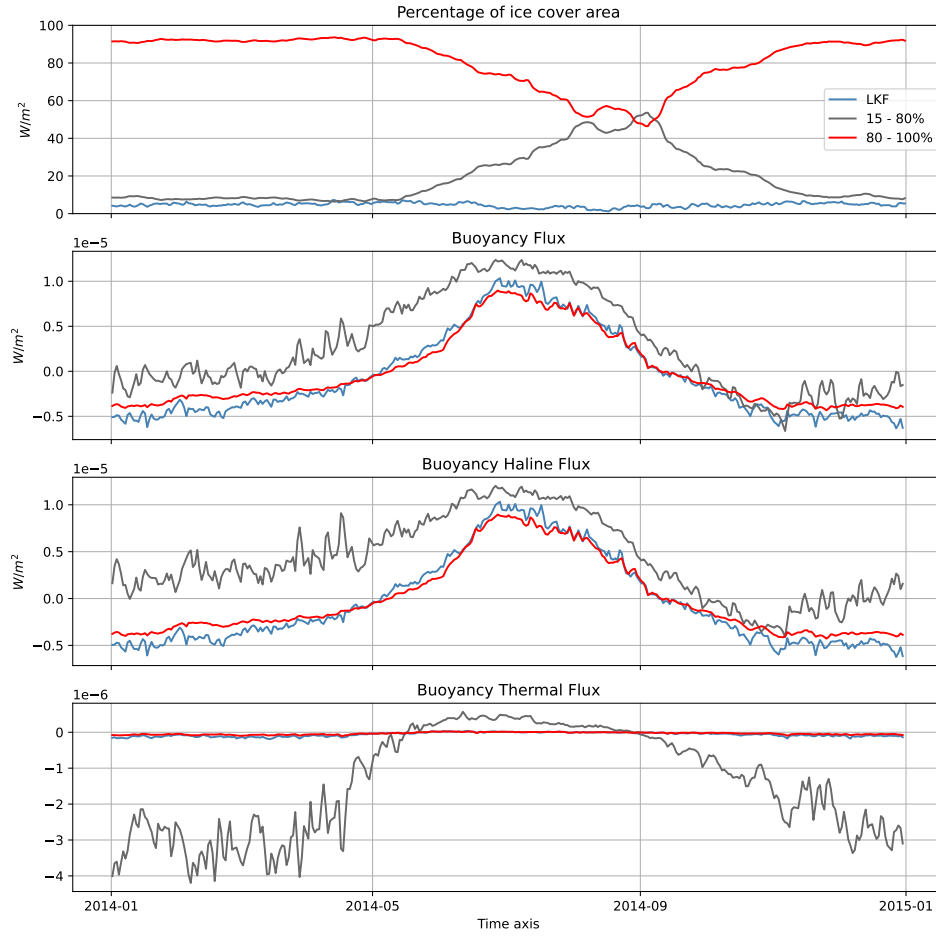


Figure 3. Time-series during 2014 of a) the area cover, b) buoyancy fluxes, c) haline buoyancy flux component, and d) thermal buoyancy flux component for linear kinematic features, transitional ice zone (ice concentration 0-80%), and pack ice (ice concentration >80%).

- TODO: Add mean over the year as a top panel.
- TODO: Map the mean transformation in SV based on the mean density of the Arctic?
- Describe seasonal cycle of water mass transformation
- Highlight the change in the magnitude
- The LKF explain around 10% of the Pack Ice WMT.
- Is it possible to estimate the effect of water masses transformation based on the outcrop of density contours in the Arctic?

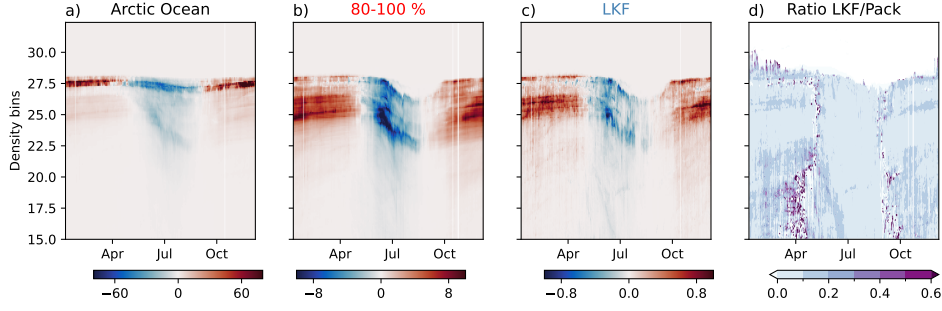


Figure 4. Hovmoller diagram of the water mass transformation (Sv). Water mass transformation of for a) the Arctic Ocean, b) Pack Ice, c) linear kinematic features, and d) ratio of water mass transformation between the LKF and the Pack Ice. Note the magnitude change in the colorbar of panels a), b), and c).

Should we do a schematic (Fig 1) as in Michaelis, J. & Lüpkes, C. but for the ocean?

2 Conclusions

reference (Bouchat et al., 2022) to argue that it's ok to use this high resolution simulation since they seem to resolve properly the LKF.

Although SEDNA reproduces well the LKF over most of the Arctic, leads missing in the Beaufort Sea could have important implications in the Arctic circulation, more research is needed here.

3 Open Research

4 Methods

Model description

Here we employ the NEMO numerical platform (release 4.0.5) to establish a $1/60^\circ$ pan-Arctic ocean-sea ice configuration named SEDNA. The ocean core (NEMO-OCE) solves the primitive equations (PE) using finite differences on an Arakawa C-grid mesh, while the ocean is coupled to the SI3 sea-ice component (NEMO-SI3). The pan-Arctic domain covers the Bering Strait on the Pacific side and extends southward to 56°N in the Subpolar North Atlantic gyre and to the Baltic Sea entrance. The ocean model incorporates a linear free surface, utilizes a 3rd order flux form scheme for momentum ad-

vection, and employs a 4th order Flux Corrected Transport (FCT) for tracer advection. Additionally, the sea-ice representation includes a 5-category model with the Elasto-Visco-Plastic (EVP) rheology activated and a land-fast capability to better simulate sea-ice behavior in specific areas.

The NCAR bulk formula from Large & Yeager (2009) to calculate air-sea fluxes and uses hourly ERA5 atmospheric data for near-surface variables. The three lateral boundaries are constrained with daily GLORYS12V1 Re-analysis data, and monthly freshwater fluxes from river discharges are combined with Greenland land ice melt data to supply SEDNA. The simulation extends over seven years (2009 to 2015) and starts from rest, utilizing initial conditions based on the World Ocean Atlas 2009 temperature/salinity and mean January 2009 ice state from PIOMAS re-analysis. Preliminary assessment against available observations and re-analyses data shows that the simulation reasonably captures key ocean and sea-ice characteristics.

LKF identification

The LKF from SEDNA were identified by using the algorithm proposed by Hutter et al. (2019). This method uses the total deformation rate defined as:

$$T_d = \left[\underbrace{\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right)^2}_{Divergence} + \underbrace{\left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right)^2 + \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)^2}_{Shear} \right]^{1/2}, \quad (1)$$

where u and v are the sea ice velocities. The deformation rate varies spatially depending on the background deformation and the sea ice properties (ρ), where the largest deformation rates occur along the boundaries of ice flows where the LKF are found. Therefore, to identify the LKFs it is required to identify the local maxima using edge detection (DoG filter) in the deformation rate field. After the edge detection, using binary map we obtain a mask with the original width of identified LKF. This mask is then used to compute the statistics presented in this study. After this step, Hutter et al. (2019) reduce the LKFs width to a single pixel to then track the LKF over time. Since our study focus on the ocean-atmosphere interactions, we only use the identification algorithm to produce a mask of LKFs during 2014 for the Arctic, where we retain the original width computed from the deformation rate. This mask is then used to quantify and contrast the impact of LKFs compared to the ice-pack.

Buoyancy flux and water mass transformation

- Describe MLD definition
- Describe buoyancy flux and decomposition between thermal and haline components.

Acknowledgments

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